

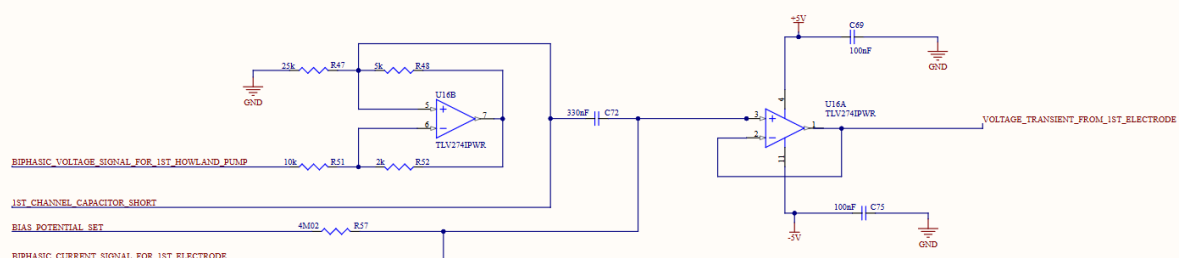
I am particularly proud of the 16-channel portable neurostimulator I designed for an NIH-funded study early in my PhD. Transitioning directly from my undergraduate studies, this served as my first major project where I managed the full lifecycle, including PCB design, firmware development, and system-level integration. Photos and design files of this system are available in my portfolio under Project 1. <https://amir1914k.github.io/Portfolio/>

1-

The component I found most fascinating was the Howland current pump. Initially, I was unfamiliar with this topology, but it proved to be an elegant solution for our hardware requirements. At its core, the Howland circuit is a specialized, op-amp-based design that converts an input voltage into a precise, stable output current. We utilized this circuit as a constant-current source because constant-voltage stimulation poses a higher risk of damaging neural tissue. By implementing a constant-current approach, we ensure that charge injection remains stable even if the impedance at the electrode-tissue interface fluctuates. While the Howland topology has inherent limitations, it provided an efficient and effective fit for our experimental application.

## Voltage-to-Current Conversion (Howland Pumps)

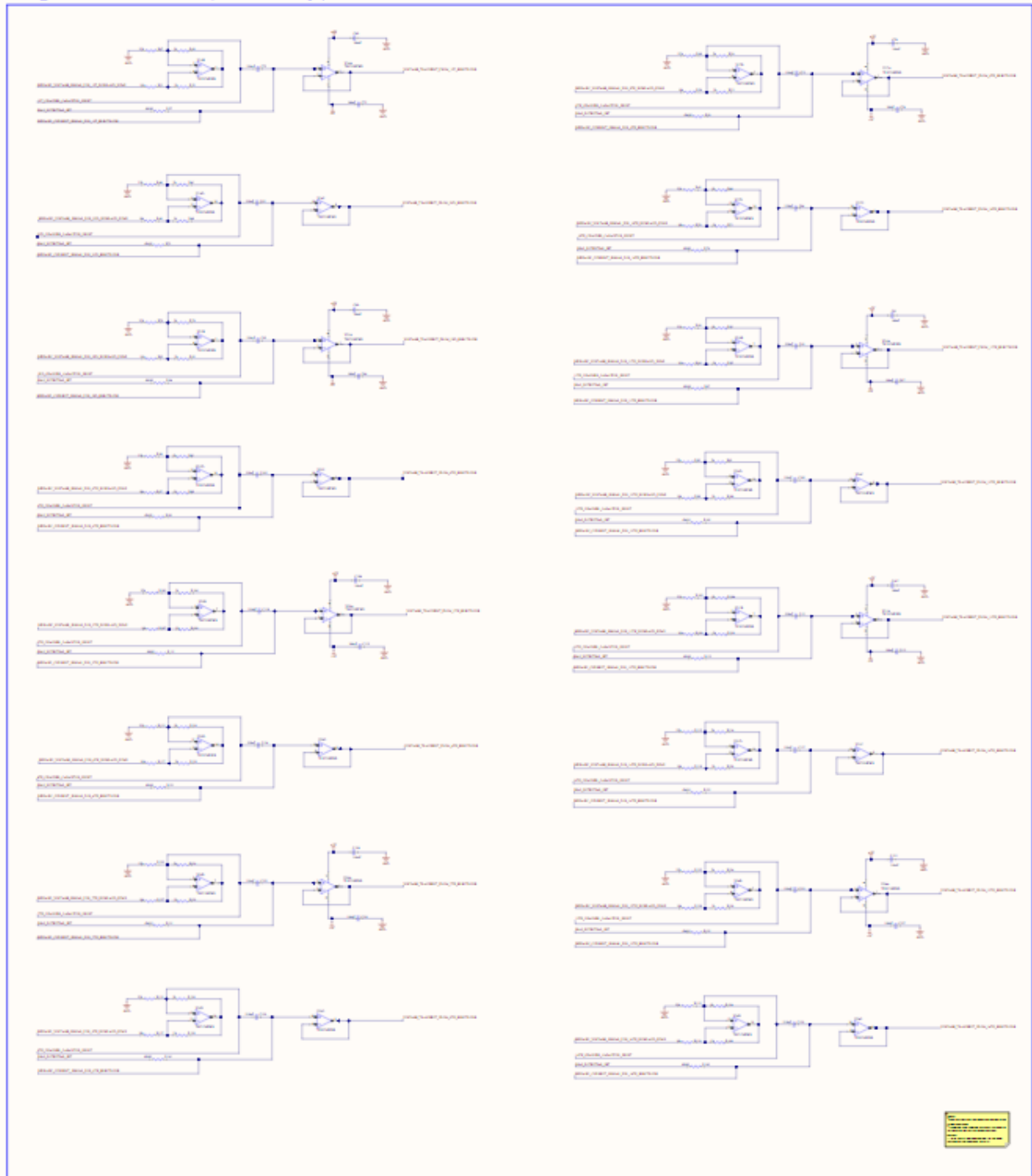
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We had 16 channels, so I had to have 16 howland-pumps!!!

#### Voltage-to-Current Conversion (Howland Pumps)



2-

To significantly improve the usability of the system, I developed a Graphical User Interface (GUI) using Python, which was a substantial upgrade over the manual, hardware-based controls used in previous iterations. Traditional systems often relied on manual analog potentiometers, which were difficult to calibrate and lacked precision. I implemented a digital, software-defined approach that allows researchers to set parameters, such as amplitude, pulse width, and anodic bias, with high, repeatable accuracy. Instead of requiring users to connect external equipment like an oscilloscope to verify electrochemical safety, I built real-time voltage transient monitoring directly into the GUI. This provides immediate visual feedback on polarization levels, which is crucial for preventing irreversible electrochemical damage to neural tissue. Furthermore, by integrating bidirectional wireless data transmission, the interface allows for stimulation adjustment without tethering the animal. This design choice significantly reduces stress for the subjects and improves signal fidelity and reproducibility during chronic studies.